

DATABASE ERD (ENTITY RELATIONSHIP DIAGRAM)

Mehmoodah Academy — School Operating System (V1)

Version: 1.0

Purpose: Define how all database tables connect before development begins.

Database: PostgreSQL

Architecture: Relational Database

SYSTEM OVERVIEW

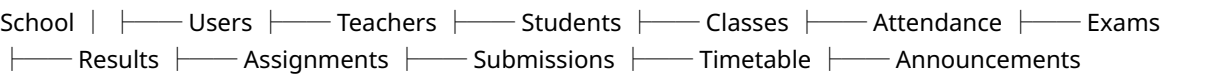


TABLE: USERS

Purpose: Authentication & Roles

Columns:

- id (PK)
- email
- password_hash
- role
- created_at
- updated_at

Relationships:

- users (1) ↓
- teachers (1)

users (1) ↓

students (1)

TABLE: TEACHERS

Purpose: Teacher Profile

Columns:

id (PK)

user_id (FK → users.id)

teacher_code

full_name

phone

subject

status

created_at

Relationships:

teacher ↓

classes

teacher ↓

attendance

teacher ↓

assignments

TABLE: STUDENTS

Purpose: Student Profile

Columns:

id (PK)

user_id (FK → users.id)

student_code

full_name

class_id (FK)

roll_number

guardian_name

phone

dob

status

created_at

Relationships:

student ↓

attendance

student ↓

results

student ↓

submissions

TABLE: CLASSES

Purpose: Class Organization

Columns:

id (PK)

name

section

teacher_id (FK)

academic_year

Relationships:

class ↓

students

class ↓

attendance

class ↓

results

class ↓

assignments

class ↓

timetable

TABLE: ATTENDANCE

Purpose: Daily Attendance

Columns:

id (PK)

student_id (FK)

class_id (FK)

teacher_id (FK)

date

status

created_at

Relationships:

attendance ↓

student

attendance ↓

teacher

attendance ↓

class

Rule:

One Attendance Per Student Per Day

UNIQUE:

student_id + date

TABLE: EXAMS

Purpose: Exam Management

Columns:

id (PK)

name

class_id (FK)

term

published

created_at

Relationships:

exam ↓

results

TABLE: RESULTS

Purpose: Store Marks

Columns:

id (PK)

exam_id (FK)

student_id (FK)

marks

percentage

grade

remarks

Relationships:

result ↓

exam

result ↓

student

Rule:

One Result Per Student Per Exam

UNIQUE:

exam_id + student_id

TABLE: ASSIGNMENTS

Purpose: Homework

Columns:

id (PK)

teacher_id (FK)

class_id (FK)

title

description

attachment_url

due_date

created_at

Relationships:

assignment ↓

submissions

TABLE: SUBMISSIONS

Purpose: Student Submission

Columns:

id (PK)

assignment_id (FK)

student_id (FK)

file_url

submitted_at

status

Relationships:

submission ↓

assignment

submission ↓

student

Rule:

One Submission Per Student

TABLE: TIMETABLE

Purpose: Class Schedule

Columns:

id (PK)

class_id (FK)

teacher_id (FK)

day

start_time

end_time

subject

Rule:

Prevent Overlap

Relationships:

class ↓

teacher

TABLE: ANNOUNCEMENTS

Purpose: Internal Notices

Columns:

id (PK)

title

body

created_by

published_at

target_role

Relationships:

announcement ↓

users

FULL RELATION MAP

users | | teachers | | classes | | attendance | | assignments | | timetable |
| | students | | attendance | | results | | submissions

classes | | students | | attendance | | exams | | assignments | | timetable

exams | | results

assignments | | submissions

INDEXES

users.email

students.student_code

teachers.teacher_code

attendance.date

results.exam_id

assignments.class_id

SOFT DELETE

Add:

deleted_at

To:

students

teachers

classes

assignments

FILE STORAGE STRUCTURE

students/

teachers/

assignments/

reports/

announcements/

END OF DATABASE ERD